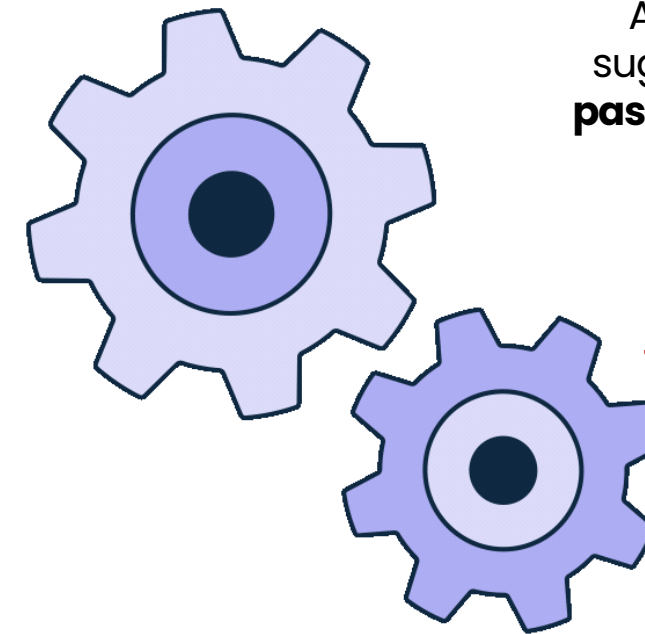


1

What is a Recommendation System?



How will you be a Long-Time customer everyday?



A recommendation engine is a system that suggests relevant items to users based on their **past behavior, preferences, or similarity** to other users/items

What to do, if this system should run 24/7?



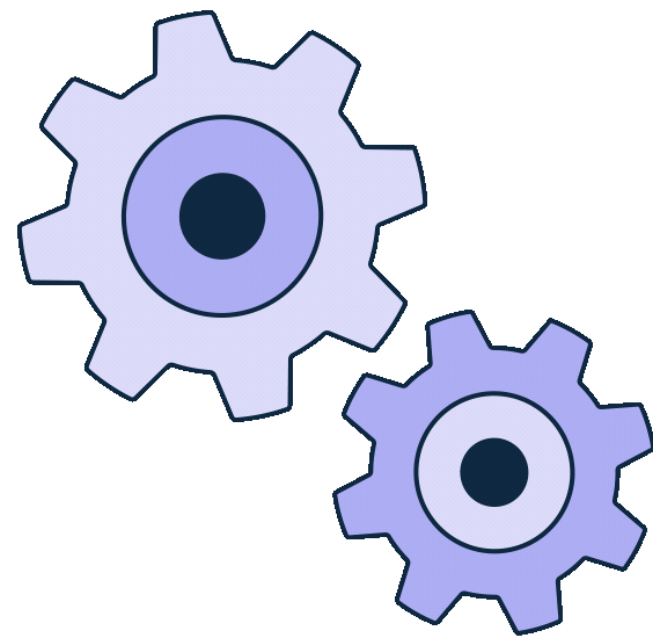
Let's understand the different ways, how we can build this System...

- The Web-Hosting Server should be **always ON**
- Behind every search or click, we should integrate this **Automatic Intelligence**.

How we are going to build in the NEXT question...

2

Different Ways to build a Recom.Eng



Recommendation
Engines

01 Popularity-based Recommend.Eng

02 Trending-Based Recommend.Eng

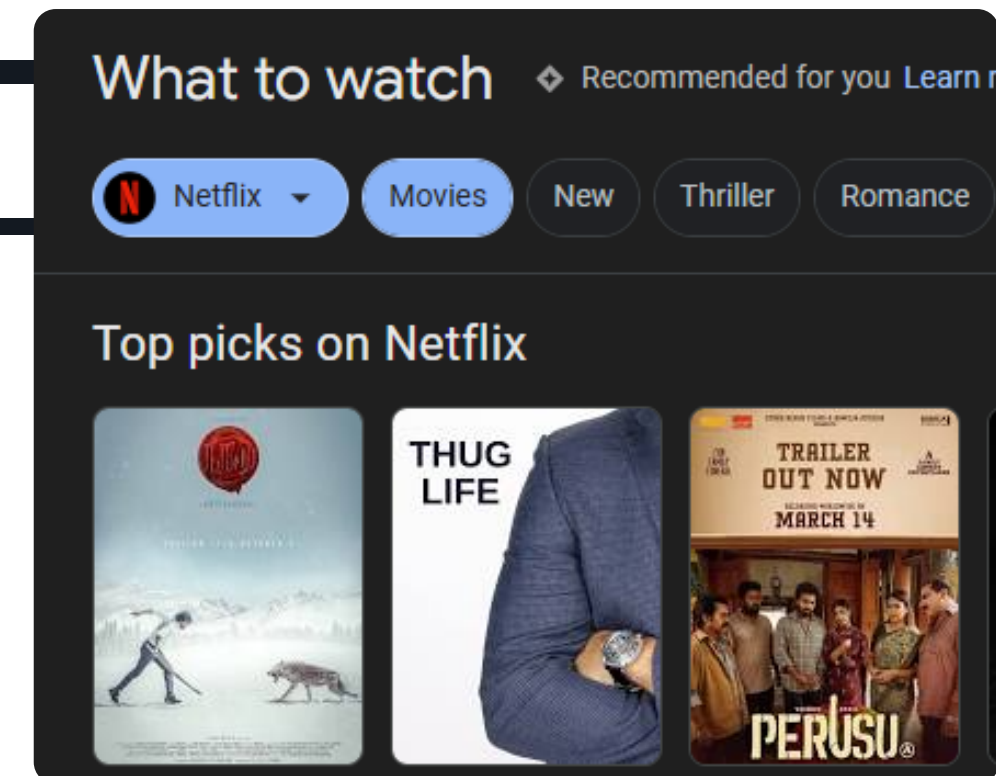
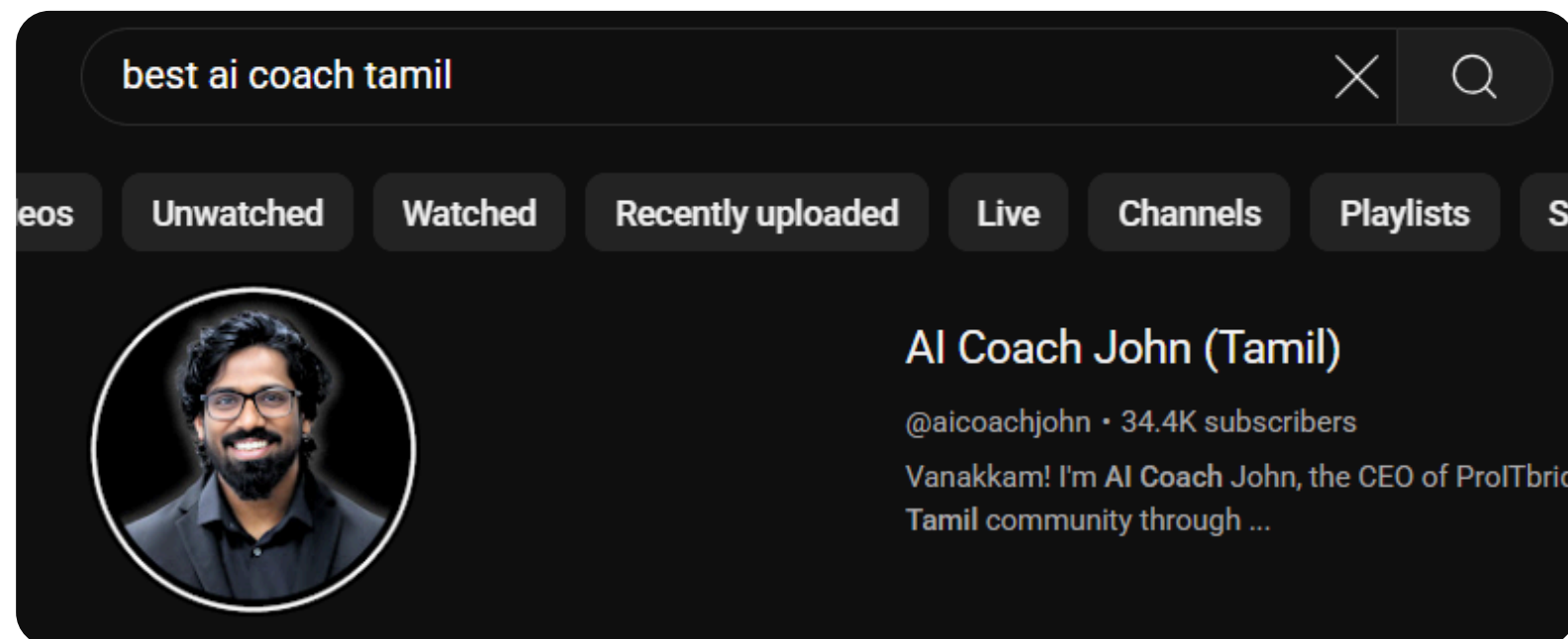
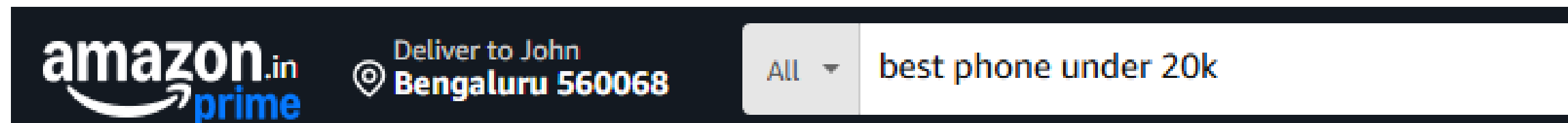
03 Content-Based Recommend.Eng

04 Collaborative-Based Recommend.Eng

05 Hybrid-based Recommend.Eng

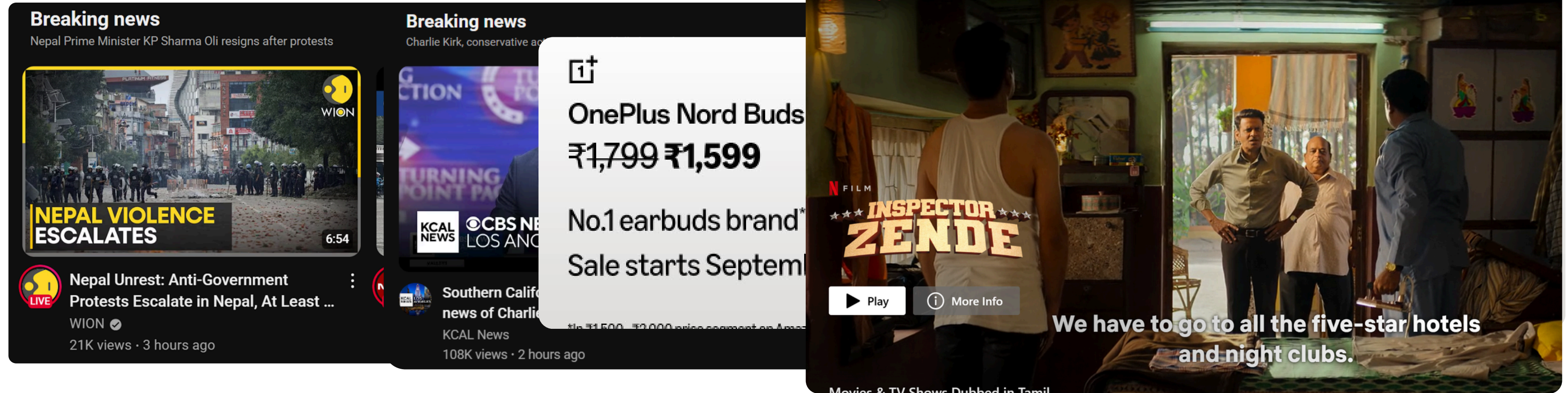
Type 1

Popularity based RE



- Suggests items that are **generally liked over time**
- Example: "Top 10 movies on Netflix," "Most sold products on Amazon".
- It's based on **aggregated long-term engagement**.

Trending based RE



- This is called **“time-based popularity”** or **“recent popularity.”**
- It's **not the same as long-term popularity** because **the item is new but suddenly spiking in views/reads.**
- It's recommended to everyone not because of historical popularity, but because **it's currently trending.**

Type 3

Content based RE

Recommends items similar to what **you already liked**, using **item features** (e.g., genres, keywords).



Genre I like

Adventure

Thriller

Comedy

Crime

It's all about Only Me...

Type 4

Collaborative based RE



UserID 1

userId	movielid	rating
1	1	4
1	5060	5
1	318	3
1	333	4



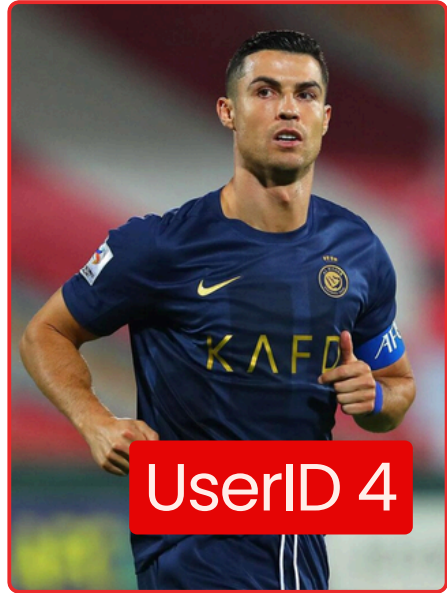
UserID 2

userId	movielid	rating
2	6	4
2	7	5
2	8	3
2	333	4



UserID 3

userId	movielid	rating
3	8	4
3	50	5
3	79	3
3	66663	4



UserID 4

userId	movielid	rating
4	653	4
4	5021	5
4	7775	3
4	333	4



UserID 345

5021



UserID 355

5060



UserID 350

7



UserID 321

318

Collaborative Based RE
This type of RE **exchanges a particular user's information between global users** and tries to find out the users with the similar interest in order to provide best recommendations.

This is how the information **exchange & recommendation** happens between these users

"UBCF"

User-to-User Based Collaborative Filtering

Type 4

Collaborative based RE



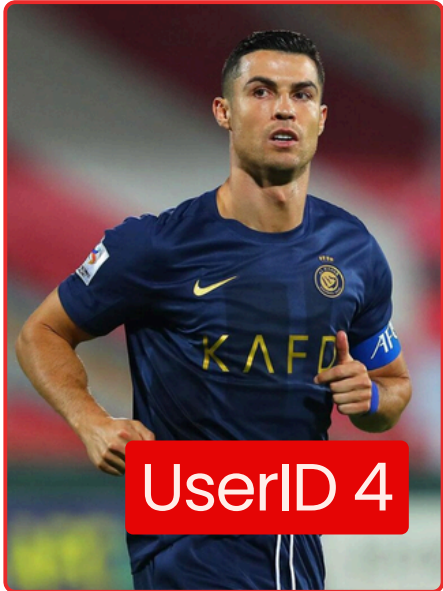
UserID 1



UserID 2



UserID 3



UserID 4

genre	movielfld	rating
Thriller	1	4
Drama	5060	5
Fantasy	318	3
Thriller	333	4

genre	movielfld	rating
Comedy	6	4
Horror	7	5
Comedy	8	3
Romance	333	4

genre	movielfld	rating
Action	8	4
Romance	50	5
Fantasy	79	3
Romance	66663	4

genre	movielfld	rating
Action	653	4
Sports	5021	5
Sports	7775	3
Thriller	333	4



UserID 345



5021



UserID 355



5060



UserID 350



7



UserID 321



318

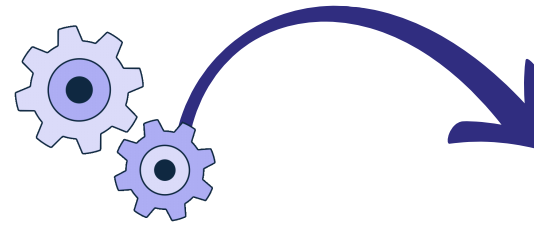
Collaborative Based RE
 This type of RE **exchanges a particular user's information between global users** and tries to find out the users with the similar interest in order to provide best recommendations.

This is how the information **exchange & recommendation** happens between these users...

"IBCF"
Item-to-Item Based Collaborative Filtering

Type 5

Hybrid based RE



Cold-Start Problem

- Happens when the system has **no data about a new user** or a new item.
- Example:** A brand-new Netflix user joins → system doesn't know what to recommend because there's **no watch history**.

userId	movieId	rating
1	1	NA
1	5060	NA
1	318	NA
1	333	NA

In Short

- Cold start** = no data at all (new user/item).
- Sparse data** = too little data spread out (lots of empty gaps).

userId	movieId	rating
1	1	4
1	5060	5
2	318	NA
2	333	NA
2	1704	4.5
3	31	NA
4	21	3
4	32	NA
4	45	NA

Hybrid Based RE

Combines **content-based + collaborative filtering** to give more accurate, balanced recommendations and reduce problems like **cold-start or sparse data**.

Sparse Data Issue

- In large systems, most users **only a few items**, so the user-item matrix is mostly empty (sparse).
- Example:** On Amazon, millions of products exist, but each user buys only a handful → **hard for collaborative filtering to find overlaps**.

A Quick Recap

- 01. Popularity-based Recommendation Engine** → Suggests the most liked or viewed items by the majority.
- 02. Trending-based Recommendation Engine** → Suggests items that are currently popular or spiking in interest.
- 03. Content-based Recommendation Engine** → Suggests items similar to what a user already liked, based on item features.
- 04. Collaborative-based Recommendation Engine** → Suggests items by finding users with similar tastes and using their choices.
- 05. Hybrid-based Recommendation Engine** → Combines multiple methods (content + collaborative, etc.) for better accuracy.